AP BIOLOGY TOPIC 6.7

Mutations, Chromosomal Alterations & Horizontal Gene Transfer

UNIT 6: GENE EXPRESSION & REGULATION (12–16% WEIGHT)

IST-2.E | IST-4.A / IST-4.B | POINT VS. FRAMESHIFT | HORIZONTAL GENE TRANSFER

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 6.7 (MUTATIONS, CHROMOSOMAL ALTERATIONS & HORIZONTAL GENE TRANSFER)

CED Core Vocabulary

What Is a Mutation? — Any permanent change in a cell's DNA nucleotide sequence. Mutations are the ultimate primary source of all new alleles and genetic variation in populations.

Endogenous Errors — Spontaneous DNA Polymerase replication errors that escape proofreading, or errors introduced during DNA repair/recombination.

Ionizing Radiation (X-rays, Gamma rays) — Generates free radicals that cause single- and double-stranded DNA backbone breaks.

Structural Rearrangements — Deletion (loss of chromosomal segment, e.g., Cri-du-chat on Chr 5), Duplication (extra gene copies that can diverge into new functions like α/β-globin families),...

Mutations in Non-Coding Regulatory Regions — Mutations in promoters, enhancers, silencers, or intron splice sites do not alter the coding triplet codons directly, but instead increase, decrease, or abolish...

UV Radiation — Causes covalent thymine dimers (T=T) that kink DNA and block replication/transcription unless fixed by nucleotide excision repair .

Chemical Mutagens & Viruses — Base analogs, intercalating agents (cause frameshifts), or viral DNA insertion.

Aneuploidy vs. Polyploidy (from Nondisjunction) — Aneuploidy = 1 extra or missing chromosome (2n±1); Polyploidy = entire extra sets of chromosomes (3n triploid, 4n tetraploid — common driver of rapid...

§2. GENE MUTATIONS: POINT (SUBSTITUTION) VS. FRAMESHIFT (INDELS)

Reading Frame Shifts

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Gene Mutations ➔ **What Is a Mutation?** ➔ **Mutations in Non-Coding Regulatory Regions**

- **What Is a Mutation?** Any permanent change in a cell's DNA nucleotide sequence. Mutations are the **ultimate primary source of all new alleles** and genetic variation in populations.

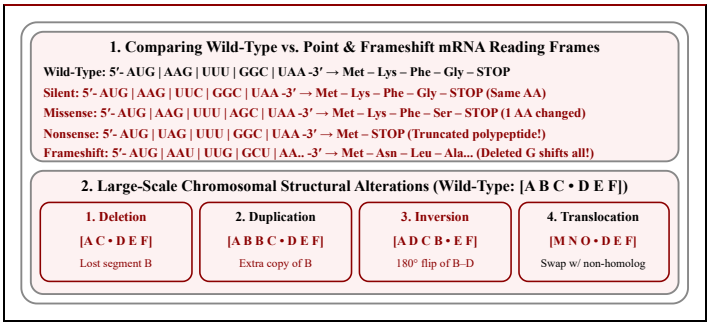
MUTATION CLASS	MOLECULAR CHANGE IN CODONS	EFFECT ON POLYPEPTIDE & PHENOTYPE
Silent (Point Substitution)	1 base substituted → new codon codes for the **exact same amino acid** (due to degeneracy at 3rd wobble base).	**No change** in primary amino acid sequence, protein folding, or phenotype.
Missense (Point Substitution)	1 base substituted → codon codes for a **different amino acid** (e.g., **GAG→GUG** Glu→Val in sickle-cell).	Impact depends on R-group chemistry & active site location: **conservative** (similar R-group = minor) vs. **non-conservative** (polar ↔ nonpolar = misfolding!).
Nonsense (Point Substitution)	1 base substituted → sense codon converted into a **premature Stop Codon (UAA, UAG, UGA)**.	Terminates translation early → produces a **truncated, shorter, usually nonfunctional polypeptide**.
Frameshift (Insertion or Deletion)	Addition or loss of nucleotides **NOT in a multiple of 3** (e.g., ±1 or ±2 bases).	**Shifts the triplet reading frame →** alters **every downstream amino acid** and usually hits an early stop codon (devastating loss of function).

- **Mutations in Non-Coding Regulatory Regions:** Mutations in **promoters, enhancers, silencers, or intron splice sites** do not alter the coding triplet codons directly, but instead **increase, decrease, or abolish the amount of protein produced** (or alter exon splicing).

[COMMON AP EXAM FRQ TRAP] **Silent Mutation ≠ "No Mutation":** A silent mutation is still a real **DNA base-pair mutation** (genotype changed); it simply has no effect on the amino acid sequence (phenotype unchanged) because of genetic code redundancy!

§4. POINT VS. FRAMESHIFT & CHROMOSOMAL MUTATIONS VISUALIZED

Codon & Chromosome Shifts



§5. CHROMOSOMAL CHANGES & PROKARYOTIC HORIZONTAL GENE TRANSFER

Polyploidy & HGT

- **Large-Scale Chromosomal Structure & Number Mutations:**
  - **Structural Rearrangements: Deletion** (loss of chromosomal segment, e.g., *Cri-du-chat* on Chr 5), **Duplication** (extra gene copies that can diverge into new functions like α/β-globin families), **Inversion** (180° reversal), and **Translocation** (segment moves between *non-homologous* chromosomes, e.g., Philadelphia chromosome in CML).
  - **Aneuploidy vs. Polyploidy (from Nondisjunction): Aneuploidy** = 1 extra or missing chromosome (**2n±1**); **Polyploidy** = entire extra sets of chromosomes (**3n** triploid, **4n** tetraploid — common driver of rapid sympatric speciation and vigorous fruit size in **plants** like wheat, strawberries, and seedless bananas!).
- **Horizontal Gene Transfer (HGT) in Prokaryotes (Variation WITHOUT Meiosis):**
  - **1. Transformation:** Bacterium takes up naked foreign DNA fragments directly from the surrounding environment and incorporates them into its genome/plasmid.
  - **2. Transduction:** Bacteriophages (viruses) accidentally package bacterial host DNA and inject it into a new bacterial host cell.
  - **3. Conjugation:** Direct cell-to-cell transfer of a plasmid (e.g., **F plasmid** or R resistance plasmid) from donor (**F^+**) to recipient (**F^-**) through a cytoplasmic **sex pilus (mating bridge)**.
  - **4. Transposition:** "Jumping genes" (**transposons**) move from one genomic location/plasmid to another via transposase.

[CORE AXIOM / REMEMBER] **HGT 3-Word Key: Transformation** = naked environmental DNA • **Transduction** = phage/virus delivery • **Conjugation** = bacterial sex pilus bridge!

§3. SOURCES OF MUTATIONS & EVOLUTIONARY IMPACT

Random Mutation, Non-Random Selection

- **Endogenous Errors:** Spontaneous DNA Polymerase replication errors that escape proofreading, or errors introduced during DNA repair/recombination.
- **Environmental Mutagens (Physical & Chemical):**
  - **UV Radiation:** Causes covalent **thymine dimers (** T=T **)** that kink DNA and block replication/transcription unless fixed by *nucleotide excision repair*.
  - **Ionizing Radiation (X-rays, Gamma rays):** Generates free radicals that cause single- and double-stranded DNA backbone breaks.
  - **Chemical Mutagens & Viruses:** Base analogs, intercalating agents (cause frameshifts), or viral DNA insertion.
- **Beneficial, Neutral, or Detrimental Depends on Environment:**
  - Whether a mutation is **beneficial** (increases fitness), **neutral**, or **detrimental** (decreases fitness) is determined by the **selective environment** (e.g., sickle-cell heterozygote advantage Hb^AHb^s in malaria regions; antibiotic resistance in hospitals).

[MECHANISTIC NOTE] **Golden Rule of Evolution on the AP Exam:** Mutations occur **completely at RANDOM** — organisms NEVER mutate "on purpose" or "in response to a need" to survive an environmental challenge! Mutation is random; **natural selection is non-random**.

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 6.7 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 6.7: Mutations, Chromosomal Alterations & Horizontal Gene Transfer ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 8 & 9 (pGLO Bacterial Transformation & Restriction Enzyme Analysis of DNA): Heat-shock transformation of E. coli with pGLO plasmid (bla ampicillin-resistance gene + GFP under arabinose-inducible araC promoter). Plate comparisons: −pGLO/LB (lawn, viability control), −pGLO/LB+Amp (no growth, antibiotic lethality control), +pGLO/LB+Amp (white colonies, transformation), +pGLO/LB+Amp+Ara (glowing green colonies under UV, inducible gene expression). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Transformation Efficiency = (Total colonies counted on selective plate) / (µg of plasmid DNA spread). Gel Electrophoresis Semi-Log Standard Curve: DNA fragment size (bp, $\log_{10}$ scale on y-axis) vs. migration distance (mm, linear scale on x-axis) yields a straight line for interpolating unknown fragment lengths. PCR amplification after n thermal cycles yields 2^n target DNA copies. **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Molecular Disruption Case Study: Splice-site mutations in the β-globin gene (β-Thalassemia) prevent the spliceosome (snRNPs) from excising an intron or cause exon skipping, shifting the mRNA reading frame and introducing a premature nonsense stop codon (UAA/UAG/UGA). In oncology, epigenetic hypermethylation of CpG islands in tumor-suppressor promoters (e.g., BRCA1 or MLH1) or histone deacetylation condenses euchromatin into transcriptionally silent heterochromatin without altering the primary DNA sequence.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 6.7 to prove why the claim holds.

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